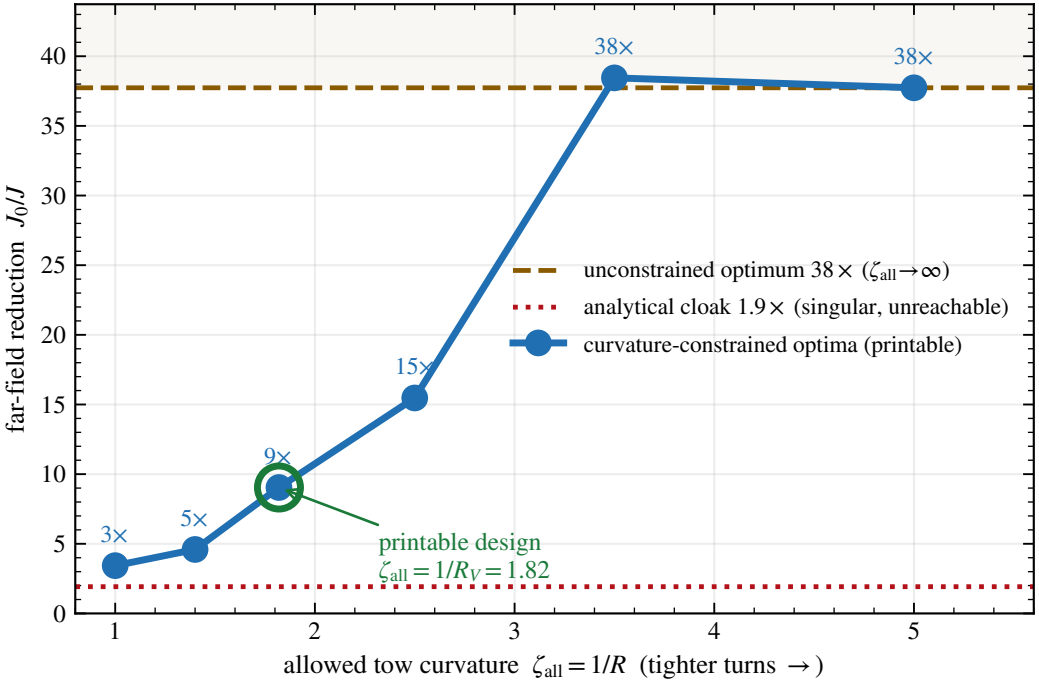
40
35
30
25
20
15
10
5
0
far-field reduction J_0/J
1
2
3
4
5
allowed tow curvature $\zeta_{all} = 1/R$ (tighter turns →)
38×
38×
15×
9×
5×
3×
printable design
$\zeta_{all} = 1/R_V = 1.82$
unconstrained optimum 38× ($\zeta_{all} \to \infty$)
analytical cloak 1.9× (singular, unreachable)
curvature-constrained optima (printable)